

README

Campus Lab Equipment Manager • Setup & Usage Guide

1. Project Overview

Campus Lab Equipment Manager is a simple Python-based application designed to manage laboratory equipment in a college campus. It helps a lab assistant or student maintain equipment records and track whether equipment is available or issued.

2. Features

- Add new laboratory equipment.
- View all equipment records.
- Search equipment by name or category.
- Issue equipment to a student.
- Return issued equipment.
- Display an equipment summary.
- Run directly in Python IDLE without external packages.

3. Technologies Used

- Python 3
- Python IDLE
- Command-line / IDLE Shell interface

No external Python libraries are required.

4. Python Concepts Used

- Variables
- Strings and integers
- Lists
- Dictionaries
- Functions

- if-elif-else statements
- for loops
- User input
- String methods
- Basic error handling

5. Project Structure

Campus-Lab-Equipment-Manager/

```

■■■ Campus_Lab_Equipment_Manager.py
■■■ README.md
■■■ Project_Report.md

```

6. Requirements

Python 3.x and Python IDLE (or another Python editor). No additional packages need to be installed.

7. How to Run the Project

Step 1	Install Python 3 if it is not already installed.
Step 2	Open Python IDLE.
Step 3	Open Campus_Lab_Equipment_Manager.py .
Step 4	Click Run → Run Module or press F5 .
Step 5	Choose an option from the displayed menu and follow the prompts.

8. Main Menu

Option	Operation
1	Add Equipment
2	View All Equipment
3	Search Equipment
4	Issue Equipment
5	Return Equipment
6	Show Summary
7	Exit

9. Example

To add equipment, select option 1 and enter the equipment name and category.

```
Enter equipment name: Digital Multimeter
Enter category: Electronics
```

```
Equipment added successfully!
Equipment ID: 1
```

10. Data Note

The current IDLE version stores equipment information in a Python list while the program is running. When the program is closed, the in-memory data is cleared.

11. Future Improvements

- Permanent file/database storage
- Student ID management
- Issue and return dates
- Administrator login
- Equipment deletion and editing
- Overdue equipment tracking
- Inventory report generation

12. GitHub Submission Reminder

Make the repository Public and submit the repository root URL. Do not submit a /tree/main or /blob/main URL.

Example: <https://github.com/USERNAME/REPOSITORY-NAME>